

## Exp. 10. Biosensor and Bio-FET Characteristics using NanoHub

### Aim

To simulate and analyse the characteristics of a BioFET-based biosensor using the NanoHUB BioSensor Lab, namely:

- To obtain the transfer characteristics ( $I_{ds}$  vs.  $V_{fg}$ ) for hybridized and unhybridized conditions of the analyte.
- To study the transient capture of target molecules by the receptor surface.
- To analyse the shift in transfer characteristics ( $I_{ds}$  vs.  $V_{fg}$ ) when the electrolyte pH is varied from 4 to 8.

### Software

NanoHub Online Tool — BioSensor Lab (Version 2).

### Theory

A BioFET (Bio-Field Effect Transistor) is a field-effect device in which the metal gate of a conventional MOSFET is replaced by an electrolyte and a receptor-functionalised sensing surface. When target biomolecules (DNA, protein or ions) present in the electrolyte bind to the immobilised receptors, they carry a net charge that modulates the surface potential of the gate insulator. This change in surface potential is electrostatically coupled into the semiconductor channel and appears directly as a shift in the threshold voltage and in the drain current of the device. The device therefore converts a biochemical binding event into a measurable electrical signal.

Two distinct conditions are compared in this experiment:

- Unhybridized case — only the single-stranded probe DNA (receptor) is present on the sensor surface. The surface charge density is low, so the channel is weakly modulated.
- Hybridized case — the complementary target DNA has bound to the probe DNA. The negatively charged phosphate backbone of the additional DNA strand increases the surface charge density, which shifts the threshold voltage and raises the drain current at the same gate bias.

The pH response of the sensor arises from protonation and deprotonation of the surface hydroxyl (–OH) groups of the gate oxide. This is described by the Nernst relation

$$E = E_0 - (2.303 RT / F) \times \text{pH}$$

which gives a theoretical maximum sensitivity of 59 mV per unit pH at 25 °C. As the electrolyte becomes more alkaline the surface loses protons, becomes more negatively charged, and the transfer curve of an n-channel device shifts towards a higher gate voltage.

The capture of target molecules is not instantaneous. It is a diffusion-limited process in which the target molecules must first travel through the electrolyte to the sensor surface before they can bind. The density of captured molecules therefore rises with incubation time and finally saturates when all available receptor sites are occupied.

### Procedure

- Open nanoHUB and log in.
- Go to Resources → Tools and select BioSensor Lab (Version 2).
- Press Launch Tool in the info section.
- In the Sensor Structure tab, select the type of sensor as ISFET / EGFET as required.
- Enter the biological parameters — analyte type (DNA), receptor density ( $1 \times 10^{12} \text{ cm}^{-2}$ ) and incubation time.

- Set the ambient temperature to 300 K.
- In the Type of Simulation to be done tab, choose the required simulation (transfer characteristics, transient capture or pH response).
- Press Simulate to view the results.
- Select Result → Transfer characteristics ( $I_{ds}$  vs.  $V_{fg}$ ) for hybridized and unhybridized cases.
- Repeat the simulation by varying the pH of the electrolyte from 4 to 8 and compare the results.

### Simulation Parameters

| Parameter                       | Value                              |
|---------------------------------|------------------------------------|
| Application                     | Biosensor Lab Version 2            |
| Sensor type                     | ISFET / EGFET                      |
| Analyte                         | DNA                                |
| Receptor density                | $1 \times 10^{12} \text{ cm}^{-2}$ |
| Incubation time                 | 60 minutes                         |
| Ambient temperature             | 300 K                              |
| Gate voltage sweep ( $V_{fg}$ ) | 0.4 V to 1.0 V                     |
| pH range                        | 4 to 8                             |

### Output — Transfer Characteristics for Hybridized and Unhybridized Cases

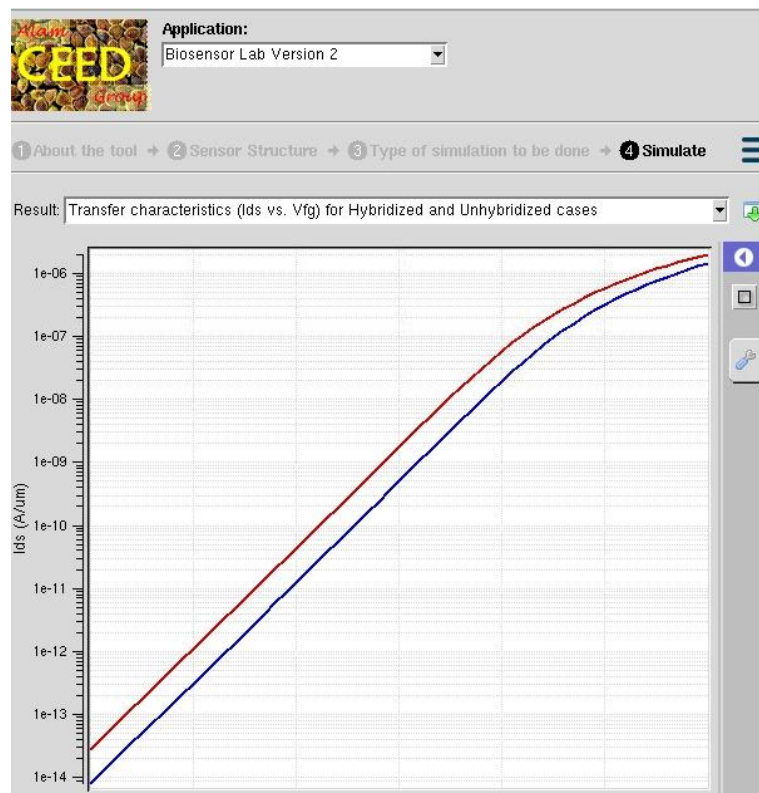


Fig. 10.1 Transfer characteristics ( $I_{ds}$  vs.  $V_{fg}$ ) for hybridized and unhybridized cases

### Explanation of Output:

The result is plotted as drain current  $I_{ds}$  (A/ $\mu\text{m}$ ) on a logarithmic scale against the front gate voltage  $V_{fg}$ . Two curves are obtained. The lower blue curve corresponds to the unhybridized case and the upper red curve corresponds to the hybridized case.

At low gate voltages both curves rise steeply and almost linearly on the log scale, which is the subthreshold region of the device. In this region the drain current increases exponentially with gate voltage. As the gate voltage increases further, both curves begin to flatten and finally approach a common saturation level of about  $1 \times 10^{-6}$  A/ $\mu\text{m}$ , which is the strong-inversion region where the channel is fully formed and the current is limited by the channel conductance rather than by the gate control.

The important observation is that the hybridized curve lies consistently above the unhybridized curve throughout the subthreshold region — the current for the hybridized case is roughly an order of magnitude higher at the same gate voltage. This is because the hybridized DNA introduces additional negative charge at the sensor surface, which effectively adds to the applied gate bias and lowers the threshold voltage of the device. The horizontal separation between the two curves is therefore the threshold voltage shift  $\Delta V_t$  produced by the hybridization event, and this shift is the actual sensing signal of the BioFET.

It is also seen that the two curves converge at high gate voltage. This is because once the device is driven into strong inversion, the current is dominated by the applied gate bias and the small additional surface charge from the DNA no longer makes a significant difference. Hence the biosensor is most sensitive when operated in the subthreshold region, and this is the region normally chosen for practical BioFET detection.

## Case Study 1: Transient Capture of Target Molecules (ISFET) — Analytical Simulation

Aim: To study how the density of captured target molecules on the sensor surface varies with incubation time for an ISFET-type biosensor.

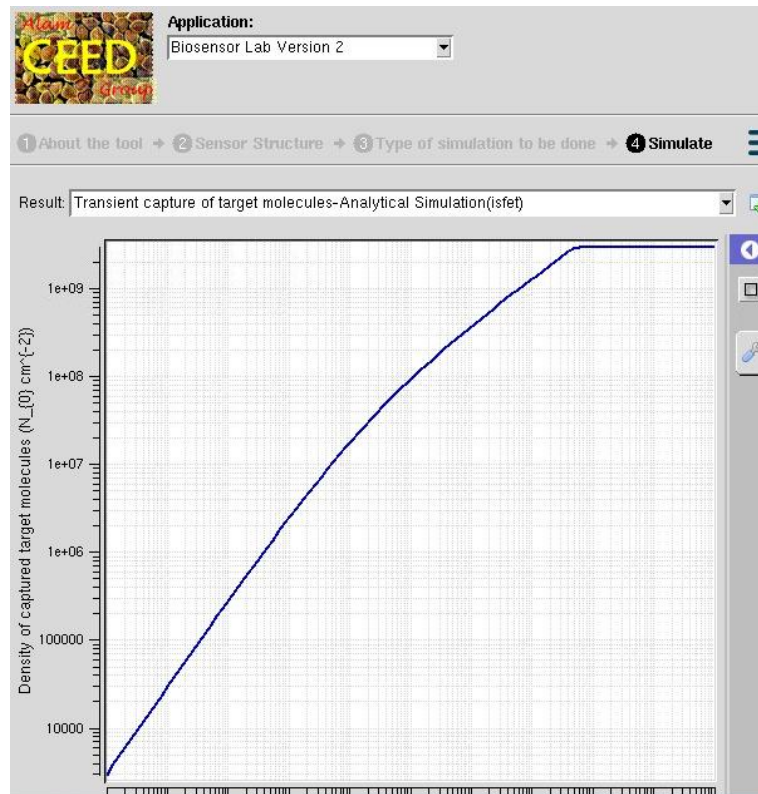


Fig. 10.2 Transient capture of target molecules — analytical simulation (ISFET)

### Explanation of Output:

The plot shows the density of captured target molecules  $N(t)$  in cm<sup>-2</sup> on a logarithmic scale against time in seconds, also on a logarithmic scale. The curve has two clearly different regions.

In the initial region the curve rises as a straight line on the log-log plot. A straight line on a log-log plot represents a power-law relationship, which confirms that the capture process in this stage is diffusion-limited. At early times the electrolyte close to the sensor surface still contains plenty of unbound target molecules, and a very large number of receptor sites are still free, so the rate of capture is governed only by how quickly the target molecules can diffuse towards the surface.

As time increases, the slope of the curve gradually reduces and the curve finally becomes a horizontal straight line at a density slightly above  $1 \times 10^9$  cm<sup>-2</sup>. This flat portion is the saturation region. Saturation occurs because almost all of the available receptor sites on the sensor surface have been occupied by target molecules, so no further binding can take place no matter how much longer the device is incubated. The region near the surface has also become depleted of free target molecules.

The practical significance of this result is that the incubation time of a biosensor must be chosen carefully. If the incubation time is too short, the sensor operates in the rising portion of the curve where the captured density — and hence the output signal — is still increasing and the reading is not reproducible. If the incubation time is long enough to reach the flat portion, the captured density is stable and the measurement becomes reliable. However, extending the incubation time beyond saturation only increases the response time of the sensor without giving any additional signal.

## Case Study 2: Transfer Characteristics ( $I_{ds}$ vs. $V_{fg}$ ) for Different pH (EGFET)

**Aim:** To study the shift in the transfer characteristics of an EGFET biosensor when the pH of the electrolyte is varied from 4 to 8, and to verify whether the response agrees with the theoretical Nernstian sensitivity of 59 mV/pH.

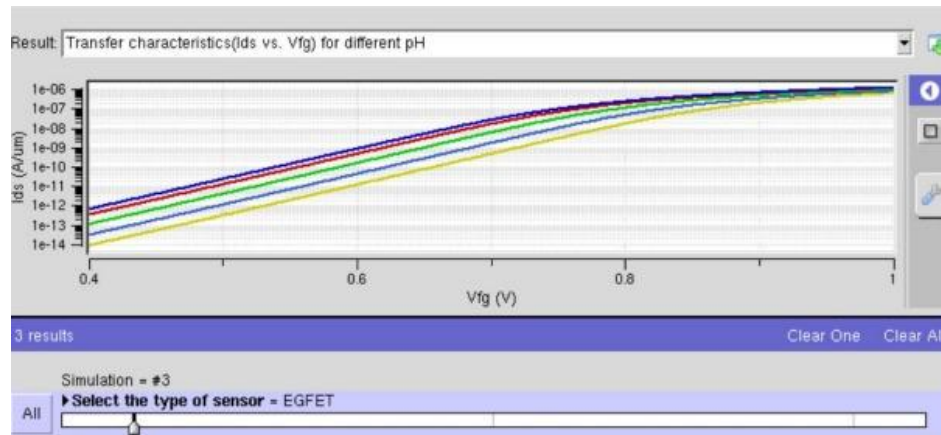


Fig. 10.3 Transfer characteristics ( $I_{ds}$  vs.  $V_{fg}$ ) for different pH values — sensor type: EGFET

### Explanation of Output:

The result shows a family of transfer curves, one for each value of pH, plotted as  $I_{ds}$  (A/ $\mu$ m) on a logarithmic scale against the front gate voltage  $V_{fg}$  over the range 0.4 V to 1.0 V. All the curves have the same characteristic shape — an exponential subthreshold rise followed by saturation near  $1 \times 10^{-6}$  A/ $\mu$ m — but they are displaced horizontally from one another.

The curves are arranged in a regular sequence: the uppermost (leftmost) curve corresponds to the lowest pH and the lowermost (rightmost) yellow curve corresponds to the highest pH. As the pH is increased, the entire curve shifts progressively towards the right, that is, towards a higher gate voltage.

This behaviour is explained by the site-binding model of the gate oxide surface. At low pH the electrolyte contains a high concentration of  $H^+$  ions, so the surface hydroxyl groups are protonated and the surface acquires a more positive charge. This positive surface charge assists in forming the channel and therefore a lower gate voltage is sufficient to turn the device on. As the pH is raised the solution becomes more alkaline, the surface groups lose protons and the surface becomes more negative, which opposes channel formation and requires a larger gate voltage to reach the same drain current. The result is the observed positive shift of the threshold voltage with increasing pH.

The most important feature of the result is that the spacing between successive curves is approximately uniform. This uniform spacing confirms that the threshold voltage varies linearly with pH, which is exactly the Nernstian behaviour predicted by theory. Since the curves shift by roughly 50–59 mV for each unit change in pH, the simulated sensitivity is close to the theoretical maximum of 59 mV/pH at 300 K.

It is again observed that the separation between the curves is largest in the subthreshold region and becomes very small once the curves saturate. This reconfirms that the device should be biased in the subthreshold region for maximum pH sensitivity.

### Result

The characteristics of the biosensor and Bio-FET were simulated using the NanoHUB BioSensor Lab. The transfer characteristics for the hybridized and unhybridized cases show a clear threshold voltage shift caused by the additional negative charge of the hybridized DNA. The transient capture simulation shows a diffusion-limited rise in the density of captured target molecules followed by saturation once the receptor sites are fully occupied. The pH-dependent transfer characteristics shift uniformly towards higher gate voltage with increasing pH, giving a near-Nernstian sensitivity of approximately 59 mV/pH. The simulation of biosensor and Bio-FET characteristics is thus accomplished.

## Exp. 11. Calculate the Potential of a pH Sensor using NanoHub

### Aim

To obtain the potential distribution of a pH sensor using the ENBIOS-1D Lab by varying the device structure — Electrolyte (E), Electrolyte/Insulator (EI) and Electrolyte/Insulator/Semiconductor (EIS) — and by varying the pH value of the electrolyte.

### Software

NanoHub Online Tool — ENBIOS-1D Lab.

### Theory

An Electrolyte–Insulator–Semiconductor (EIS) sensor is a potentiometric sensor built from three functional regions:

- An electrolyte, which contains the chemical species ( $H^+$  and  $OH^-$  ions) whose concentration is to be measured.
- An insulator, which allows field-effect coupling between the electrolyte and the semiconductor without permitting any leakage current between them.
- A semiconductor, in which the electrostatic changes produced by the electrolyte are registered as a change in the surface potential and the width of the depletion region.

The operation of the sensor is based on the site-binding model. The surface of the gate insulator carries amphoteric hydroxyl ( $-OH$ ) groups which can either accept a proton to become positively charged ( $-OH_2^+$ ) or donate a proton to become negatively charged ( $-O^-$ ), depending on the  $H^+$  concentration of the solution. The resulting surface charge sets up an interface potential given by the Nernst relation

$$\psi = \psi_0 - 2.303 (kT/q) \times pH$$

which corresponds to a maximum theoretical sensitivity of 59 mV per unit pH at room temperature.

Charge neutrality in the electrolyte is maintained by the formation of an electrical double layer at the interface. The characteristic screening length of this double layer is the Debye length, given by  $\lambda_D = \sqrt{(\epsilon kT / 2q^2 n_0)}$ . Because the Debye length in an aqueous electrolyte of moderate ionic strength is only of the order of a few nanometres, the entire potential drop in the electrolyte occurs over an extremely short distance immediately adjacent to the insulator, and the bulk of the electrolyte remains field-free at a constant potential.

The ENBIOS-1D tool solves the one-dimensional Poisson–Boltzmann equation across the whole structure and returns the DC potential as a function of distance, which allows the potential drop across each individual region to be identified.

### Procedure

- Open nanoHUB and log in.
- Go to Resources → Tools and select ENBIOS-1D Lab.
- Press Launch Tool in the info section.
- From the Structure drop-down menu, select the required structure — Electrolyte (E), Electrolyte/Insulator (EI) or Electrolyte/Insulator/Semiconductor (EIS).
- In the Device tab, set the semiconductor material, doping type, insulator and electrode parameters.
- Open the Electrolyte tab and enter the electrolyte thickness, solvent, pH simulation method and the pH-defining solutes.
- Press Simulate to run the simulation.
- Select Result → Potential (DC) to view the potential profile.
- Change the pH value of the electrolyte and repeat the simulation.

- Change the insulator thickness and length, repeat the simulation, and compare all the results.

### Simulation Parameters

| Parameter                                     | Value                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------|
| Semiconductor material                        | Si                                                                     |
| Relative permittivity                         | 11.68                                                                  |
| Electron mobility                             | 1417 cm <sup>2</sup> /V·s                                              |
| Hole mobility                                 | 470.5 cm <sup>2</sup> /V·s                                             |
| Intrinsic carrier concentration               | $1.239 \times 10^9$ /cm <sup>3</sup>                                   |
| Doping type                                   | n-type                                                                 |
| Electrode 1 type                              | Ideally polarizable                                                    |
| Electrode 2 type                              | Faradaic                                                               |
| Electrolyte thickness                         | 1 μm                                                                   |
| Solvent                                       | Water                                                                  |
| pH simulation method                          | Explicit (pH = 7)                                                      |
| pH defining solute                            | H <sup>+</sup> Cl <sup>-</sup> (strong acid), $1 \times 10^{-8}$ mol/L |
| Na <sup>+</sup> Cl <sup>-</sup> concentration | 0.001 mol/L                                                            |
| Result selected                               | Potential (DC), DC bias step = 1                                       |

## Output — Electrolyte/Insulator/Semiconductor (EIS) Structure

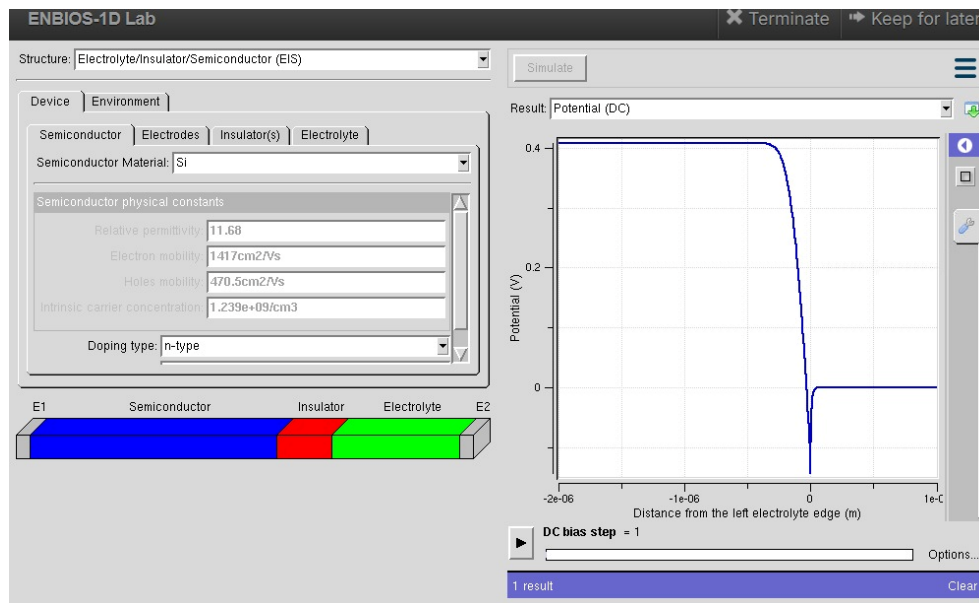


Fig. 11.1 Potential (DC) profile for the EIS structure with n-type silicon

### Explanation of Output:

The device schematic at the bottom of the window shows the complete EIS stack: electrode E1, the silicon semiconductor (blue), the insulator (red), the electrolyte (green) and electrode E2. The plot on the right shows the DC potential in volts against the distance from the left electrolyte edge, over the range  $-2 \times 10^{-6}$  m to  $1 \times 10^{-6}$  m.

The potential curve has three clearly distinguishable regions. Inside the semiconductor, from about  $-2 \times 10^{-6}$  m up to nearly the origin, the potential is flat and constant at approximately 0.4 V. This is the neutral bulk of the n-type silicon, where the free electron concentration is high enough to screen out any external field completely, so no potential gradient can be sustained.

Very close to the origin the curve falls extremely steeply from 0.4 V down to 0 V. This sharp drop occurs across the thin insulator layer and the semiconductor depletion region. Since the insulator contains no mobile charge carriers, the entire applied bias must be dropped across it, which produces this very high electric field over a very short distance. A small undershoot is seen at the bottom of the drop, indicating a slight accumulation of charge at the insulator–electrolyte interface.

Beyond the origin, on the electrolyte side, the potential immediately recovers to 0 V and remains flat. This is because the mobile ions in the electrolyte redistribute themselves to form an electrical double layer that screens the field within a distance of only a few Debye lengths. On the scale of this plot, that screening distance is so small that the electrolyte appears completely field-free.

The overall conclusion from this plot is that in an EIS pH sensor almost the whole potential drop is confined to the insulator and the adjacent depletion and double-layer regions. Any change in the surface charge caused by a change in pH therefore acts directly on this narrow region and produces a measurable shift in the semiconductor surface potential — which is precisely the mechanism used for pH sensing.

## Case Study 1: Potential Profile of the Electrolyte/Insulator (EI) Structure

Aim: To obtain the potential distribution when the semiconductor region is removed and the structure is reduced to Electrolyte/Insulator (EI), and to compare it with the complete EIS structure.

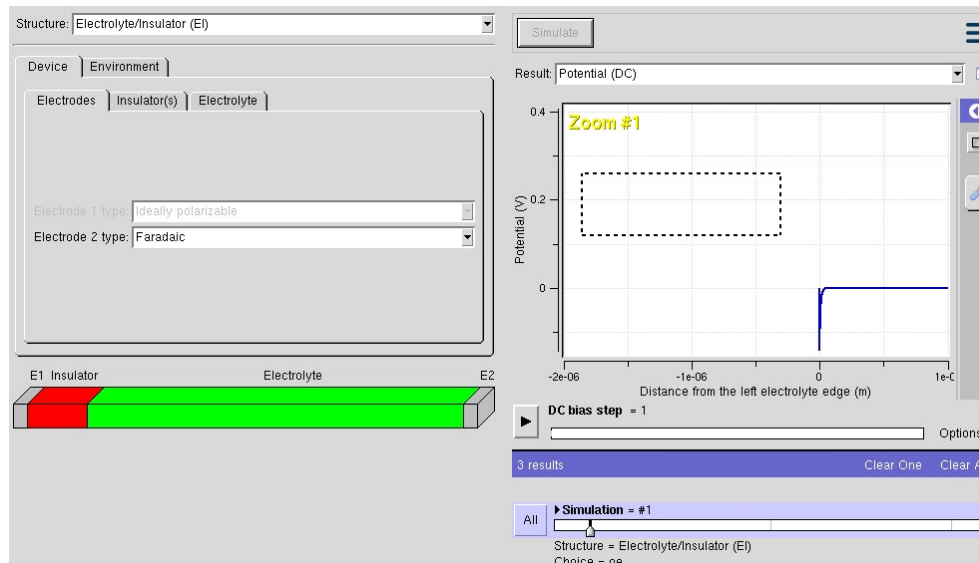


Fig. 11.2 Potential (DC) profile for the Electrolyte/Insulator (EI) structure

### Explanation of Output:

In this case the structure consists only of electrode E1, the insulator (red) and the electrolyte (green) terminated by electrode E2. Electrode 1 is set as ideally polarizable, which means no charge transfer takes place across it, while electrode 2 is set as Faradaic, which allows charge transfer and therefore fixes the reference potential of the electrolyte.

The resulting potential profile is markedly different from the EIS case. The high plateau at 0.4 V has completely disappeared. Instead, the potential remains essentially at 0 V across the whole plotted region, with only a very small localised dip occurring right at the origin, where the insulator meets the electrolyte.

This result is expected because in the absence of a semiconductor there is no depletion region and no semiconductor surface potential to be developed. The only remaining feature is the small potential step that forms in the electrical double layer at the insulator–electrolyte interface. The zoom box drawn on the plot indicates the region that would need to be magnified to resolve this feature, since on the present scale it is almost invisible.

The comparison between Fig. 11.1 and Fig. 11.2 makes the role of the semiconductor very clear. The semiconductor is what converts the small interfacial potential change produced by the electrolyte into a large, easily measurable band bending. Without it, the structure behaves simply as a capacitor with a very small interfacial potential and provides almost no useful sensing signal.

## Case Study 2: Potential Profile of the Electrolyte-only (E) Structure

**Aim:** To obtain the potential distribution when only the electrolyte region is present, and to identify the reference condition of the sensor.

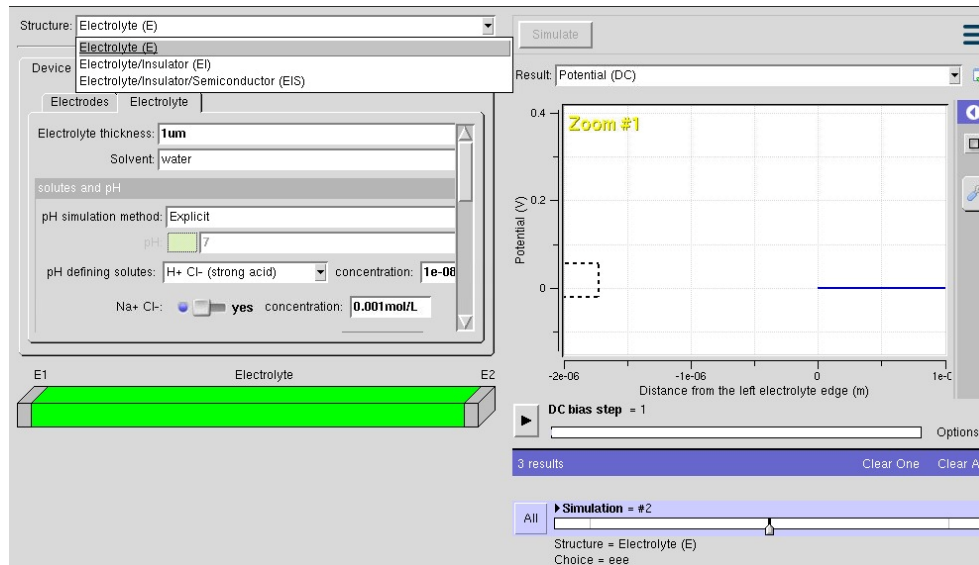


Fig. 11.3 Potential (DC) profile for the Electrolyte-only (E) structure, pH = 7

### Explanation of Output:

Here the structure is reduced to a single electrolyte region of thickness 1  $\mu\text{m}$  placed between the two electrodes. The solvent is water, the pH simulation method is set to Explicit with pH = 7, the pH-defining solute is  $\text{H}^+ \text{Cl}^-$  (strong acid) at a concentration of  $1 \times 10^{-8} \text{ mol/L}$ , and a supporting electrolyte of  $\text{Na}^+ \text{Cl}^-$  at 0.001 mol/L is included.

The resulting potential is a perfectly flat horizontal line at 0 V across the entire electrolyte region. There is no gradient anywhere in the plot.

This flat response is the expected and correct result. The electrolyte is an ionic conductor: any electric field that tries to establish itself inside it is immediately screened by the redistribution of the mobile  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{H}^+$  and  $\text{OH}^-$  ions. This screening takes place within a distance of the order of the Debye length, which for the given ionic strength of 0.001 mol/L is only a few nanometres — far smaller than the 1  $\mu\text{m}$  resolution of this plot. The bulk of the solution therefore remains completely field-free and at a uniform potential.

The presence of the supporting  $\text{Na}^+ \text{Cl}^-$  electrolyte is important here. It sets the ionic strength of the solution and hence controls the Debye screening length. A higher supporting electrolyte concentration gives a shorter Debye length and stronger screening, which is why the potential appears perfectly flat in this simulation.

This case therefore establishes the baseline condition of the sensor. Any potential variation observed in the EI and EIS structures must originate at the interfaces, not from the bulk electrolyte itself.

### Comparison of the Three Structures

| Structure                  | Region carrying the potential drop      | Observed potential profile                                        | Sensing capability                   |
|----------------------------|-----------------------------------------|-------------------------------------------------------------------|--------------------------------------|
| E (Electrolyte only)       | None — bulk is screened by mobile ions  | Flat line at 0 V throughout                                       | None; reference condition only       |
| EI (Electrolyte/Insulator) | Insulator–electrolyte double layer only | $\approx 0 \text{ V}$ with a small localised dip at the interface | Very weak; no band bending available |

|                                              |                                                    |                                                                                 |                                   |
|----------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------|
| EIS<br>(Electrolyte/Insulator/Semiconductor) | Insulator and<br>semiconductor depletion<br>region | Flat 0.4 V in Si, sharp drop<br>to 0 V at the interface, flat<br>in electrolyte | Strong; full pH<br>sensing action |
|----------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------|

## Analytical

Calculation of the theoretical pH sensitivity of the sensor at  $T = 300$  K.

The interface potential of the sensor is governed by the Nernst relation:

$$\psi = \psi_0 - 2.303 (kT/q) \times \text{pH}$$

Given:  $k = 1.38 \times 10^{-23}$  J/K,  $T = 300$  K,  $q = 1.6 \times 10^{-19}$  C.

The thermal voltage is

$$kT/q = (1.38 \times 10^{-23} \times 300) / (1.6 \times 10^{-19}) = 0.02587 \text{ V} \approx 25.87 \text{ mV}$$

Therefore the sensitivity per unit pH is

$$\Delta\psi / \Delta\text{pH} = 2.303 \times 25.87 \text{ mV} = 59.57 \text{ mV/pH} \approx 59 \text{ mV/pH}$$

Hence the maximum theoretical (Nernstian) sensitivity of the pH sensor at 300 K is approximately 59 mV per unit pH. For a change of electrolyte pH from 4 to 8, the total expected shift in surface potential is

$$\Delta\psi = 4 \times 59 \text{ mV} = 236 \text{ mV} \approx 0.236 \text{ V}$$

In a practical device the measured sensitivity is usually slightly lower than this value. This is because the surface hydroxyl groups of the insulator have a finite site density and a finite intrinsic buffer capacity, so the surface potential does not respond fully to every change in  $\text{H}^+$  concentration. Insulators with a high site density such as  $\text{Ta}_2\text{O}_5$  and  $\text{Al}_2\text{O}_3$  therefore give a response much closer to the ideal 59 mV/pH than  $\text{SiO}_2$  does.

## Result

The potential of the pH sensor was obtained using the ENBIOS-1D Lab for the Electrolyte, Electrolyte/Insulator and Electrolyte/Insulator/Semiconductor structures. The electrolyte-only structure shows a completely flat potential because the field is screened by the mobile ions within a few Debye lengths. The EI structure shows only a small potential step at the insulator–electrolyte interface. The complete EIS structure shows a constant potential of 0.4 V in the n-type silicon bulk followed by a sharp drop to 0 V across the insulator and depletion region, confirming that the entire potential drop is confined to the interface region where sensing takes place. The theoretical Nernstian sensitivity was calculated as approximately 59 mV/pH at 300 K. The calculation of the potential of a pH sensor using NanoHub is thus accomplished.